

Location Decision of Battery Swapping Stations Combined with Battery Swapping Warehouses

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Abstract—The worldwide promotion of Electric Vehicles (EVs) is extremely important to reduce the use of fossil fuels and thus achieve the goal of reducing carbon emissions. The layout and application of EV replenishment facilities largely influence consumers' willingness to purchase EVs, thus affecting the in-depth promotion of new energy vehicles. At the same time, the large number of charging and swapping stations connected to the power grid also has an impact on the existing power system. In this paper, we propose a strategy for the layout of charging/swapping stations, which can be used to promote the development of new energy EV charging modes. Specifically, we propose a model combining charging/swapping stations and swapping warehouses, which optimizes the location and the number of swapping warehouses, enabling EV owners to reach the swapping stations faster, increasing the service coverage, and fulfilling the demand while saving costs and increasing operators' revenue. Since the battery swapping warehouses are not connected to the power grid, the construction cost and operation cost are much lower than that of the swapping stations. The used batteries are transported back to the charging/swapping station to replenish energy during the low-load hours of the grid, which completes the peak-shaving and valley-filling of the grid load.

Keywords—Electric vehicles, Swapping stations, Genetic Algorithm, Facilities location, Warehouse and inventory

I. INTRODUCTION

Compared with fuel cars, the efficiency of energy replenishment has become an important consideration for users of new energy vehicles. After 10 years of development, a large number of charging piles have been put into use in China. While a conventional fuel car can be easily refueled in minutes at an accessible gas station, Electric Vehicles (EVs) [1] require specific equipment and significant charging time. It always takes 3-6 hours to fully charge, and it even takes 30 minutes to take a fast charge [2]. Then regarding range anxiety, EVs are usually limited to 500km in a single trip with a full battery, and consumers also worry about the high battery cost.

In contrast to the charging mode, the power swapping mode can effectively solve the users' mileage anxiety, improve the driver's income, eliminate the hidden danger of the battery, and reduce the initial purchase cost. Battery Swapping Stations (BSSs) are one of the main types of EV refueling facilities. By building enough swapping stations, when an EV arrives at a swapping station, a third-party company will directly unload a battery that is out of charge and replace it

with a fully charged battery so that the EV can continue to travel through the battery rapid exchange system [3]. The removed batteries can then be charged in a separate mode in a charging facility, thus decoupling the battery from the EV during refueling.

The feature of the battery as an energy storage device brings an obvious shortcoming, that is, the battery charging time is significantly longer than the refueling time of a traditional fuel car. However, in the power swapping mode, the layout of the new energy EV replenishment facilities can also form a significant advantage by using the feature that the battery can be flexibly plug-in and out. Traditional fuel cars need to go to the gas station to replenish energy. Gas stations need to be equipped with complete storage tanks, facility footprint, refueling facilities, and so on, which lead to large total costs. Furthermore, it may result in traffic congestion during peak periods since the vehicle queueing time is too long [4]. On the other hand, if we want to improve the development of the power swapping mode, enough battery swapping stations have to be built, but the construction cost of battery swapping stations is also quite high, and it is easy to fall into the dilemma of "chicken or egg first" [5]. Luckily, the mature development of the storage and supply chain allows us to optimize and solve this problem by using the characteristics of removable EV batteries. Since the cost of batteries is relatively high, the demand in each region is relatively stable, and EV users have certain requirements for the efficiency of reaching the swapping station [6]. This paper proposes a battery swapping system using area battery swapping warehouses, which optimizes the number of batteries by reasonably distributing the battery inventory in each battery swapping warehouse and timely dispatching them to achieve the goal of satisfying more customers' demands and increasing the total profit of the region. This mode will also have an impact on the power grid, since the battery swapping warehouses are not connected to the power grid, it will not affect the power distribution of the grid nodes, but the used batteries received from users in the swapping warehouse will be transported back to the charging and swapping site for charging during the low-load period, which is helpful to achieve the effect of peak shaving and valley filling.

The remaining paper is organized as follows. Section II is the problem setup, in which we construct the framework of a new battery swapping mode. Then in Section III, we set up the models that are needed in solving this problem including the

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site selection model, the station operation model, and the inventory control model. In Section IV, we simulate the new battery swapping mode to show the effectiveness of our swapping facilities layout in increasing service rates, peak shaving and valley filling of grid loads. Finally, Section V concludes this paper.

II. PROBLEM SETUP

In the conventional battery swapping mode (Fig. 1), the BSSs need to be equipped with adequate charging piles, energy storage facilities, a large footprint, and so on, hence the total cost is large. The large construction cost results in limited battery swapping stations to fulfill demands. In addition, the long vehicle queues may cause traffic congestion during rush hour. Therefore, we propose the concept of Battery Swapping Warehouse (BSW) and Center Charging Stations (CCS).

Battery Swapping Warehouses: It refers to the establishment of more small battery warehouses around the regional center swapping/charging stations, built by EV operators based on the regional center swapping stations, to assist in the EV swapping behavior. These small depots, or BSWs, are not connected to the grid and have flexible siting procedures, small footprints and low hardware requirements.

Center Charging Stations: In addition to providing charging and swapping services to EV users who arrive at the stations, they provide slow charging of batteries transported back from the swapping warehouses during low-load periods, playing a role in peak shaving and valley filling for the grid. Relying on a distribution network, they deliver fully charged batteries to the swapping warehouses at a regular frequency.

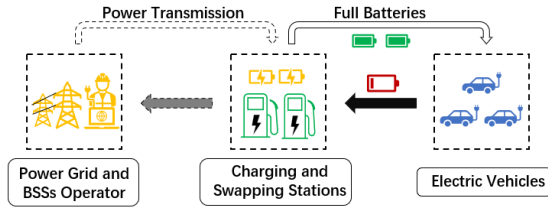


Fig. 1. The conventional battery swapping mode.

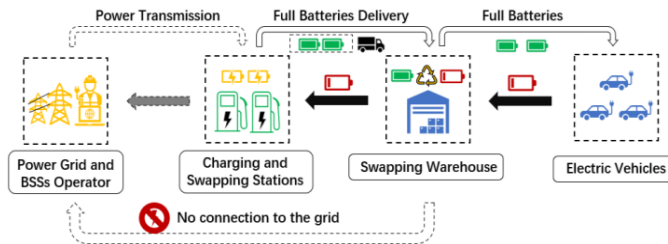


Fig. 2. The swapping mode with battery swapping warehouses.

The layout structure of the conventional swapping mode with swapping warehouses proposed in this paper is shown in Fig. 2. Compared with the conventional swapping mode, the new mode has some advantages and challenges to address:

Advantages of BSS+BSW mode: Convenience, increased service coverage, increased revenue, and the BSS plays a role in peak shaving and valley filling, with less energy loss of the battery.

Challenges of BSS+BSW mode: increased rental costs, operating costs, and increased transportation costs.

III. MODELS

To address the above challenges, we consider three parts of modeling. The first is the site selection model, which increases the coverage of the battery swapping service and shortens the time for customers to reach the BSSs by optimizing the layout of the number and location of the swapping depots. The second is the station operation model, where the replaced batteries are connected to the grid for charging and will have an impact on the grid. From the perspective of the BSSs operator and the grid side, it can save electricity and increase profitability by reasonably transporting batteries to centralized charging stations and adjusting battery charging time; it can also adjust electricity load and shave peak power. The third is the inventory model. Due to the high cost, it is not suitable to prepare too many storages, which lead to high inventory costs, so it is necessary to clearly model the battery inventory of BSSs/BSWs.

A. Site Selection Model

To begin with, select candidate sites for the BSS/BSW in the region based on the restrictions of distance and demand to ensure coverage and service rates [7].

Firstly, identify the candidate site. Assume that the region map is abstracted as a coordinate system, using horizontal and vertical coordinates to represent relative positions. Denote $Z_l(x, y)$ as a node position, $l = 1, 2, \dots, m$. Denote $D_k(x, y)$, $k = 1, 2, \dots, u$ as a node of demand points that are converted from the traffic flow through this node, and c_k indicates the demand value. There are demand points in the neighborhood of Z_l , and the neighborhood is a small area with the radius set as $Th_{distance}$. Then set a threshold Th_{demand} : if the total demand in the neighborhood of Z_l is below Th_{demand} , it is not possible to make a profit by operating a swapping station here. If the demand summation is no less than Th_{demand} , this point satisfies the condition of being a candidate point. $d_{lk} = \|D_k - Z_l\|_2$ indicates the distance between the demand point D_k and Z_l . For every sites l , if it meets the condition of

$$\sum_{k: d_{lk} \leq Th_{distance}} c_k \geq Th_{demand}, \text{ it is selected as a candidate site.}$$

Secondly, select q BSWs from the candidate sites. Consider a set $I = \{1, 2, \dots, n\}$ of candidate sites for facility location, and a set $J = \{1, 2, \dots, u\}$ of demand points. Each facility $i \in I$ has a fixed cost f_i . ω indicates the unit transportation cost; d_{ij}

indicates the distance from i to j , c_j indicates the demand of node j . Then the objective function of the model is:

$$\min_{x,y} Z = \sum_{i=1}^n \sum_{j=1}^u \omega d_{ij} x_{ij} c_j + \sum_{i=1}^n f_i y_i \quad (1)$$

Decision variables:

$$y_i = \begin{cases} 1, & \text{indicates that the } i^{\text{th}} \text{ BSS / BSW is open} \\ 0, & \text{else} \end{cases} \quad (2)$$

$$x_{ij} = \begin{cases} 1, & \text{indicates that the } j^{\text{th}} \text{ demand point is} \\ & \text{served by the } i^{\text{th}} \text{ BSS/BSW station} \\ 0, & \text{else} \end{cases} \quad (3)$$

Constraints:

$$\sum_{i=1}^n y_i = q \quad (4)$$

$$\sum_{i=1}^n x_{ij} = 1, \forall j = 1, 2, \dots, u \quad (5)$$

$$d_{ij} = \|D_j - Z_i\|_2 \leq R \quad (6)$$

$$\sum_{i=1}^n c_j x_{ij} \leq Q_i^*, \forall j = 1, 2, \dots, u \quad (7)$$

$$x_{ij} \leq y_i, \forall i = 1, 2, \dots, n; \forall j = 1, 2, \dots, u. \quad (8)$$

Eq. (4) means to select q battery swapping warehouses, and y_i is a 0-1 variable: if site i is chose to build a battery swapping warehouse, $y_i = 1$, otherwise, $y_i = 0$. Eq. (5) represents the demand distribution relationship between demand points and BSWs, and ensures that the locations without facilities do not have demand points to serve. Eq. (6) indicates the distance from i to j no longer than the coverage radius R . Eq. (7) constraints the capacity of the battery swapping warehouse, with the Q_i^* the maximum capacity of station i , which is derived from inventory level model later. Eq. (8) means that the demand point can be served only after the BSW is established at the candidate point y_i .

Genetic Algorithm (GA) are used to solve the problem because of the large number of demand points in a city and the uncertainty of the number of BSS/BSW. It is a computational model of biological evolutionary process that simulates the mechanism of natural selection and genetics of Darwinian biological evolution, and is a method to search for the optimal solution by simulating the natural evolutionary process. The whole process of the algorithm is to start from any initial population and make the population proceed to the best region in space from one generation to the next by random selection, crossover and mutation, until the optimal point is reached.

B. Station Operation Model

Since the BSWs are not connected to the power grid, the used batteries unloaded at the BSWs can be transported back to the centralized charging station and connected to the grid

for charging in an orderly manner during low-load periods, where the station operation model is used to optimize the load of the grid.

The goal is to reduce the total cost and increase revenue of BSSs/BSWs operators. The total cost includes the gap between the batteries discharging income $f^D(P_t^D)$ and charging cost $f^C(P_t^C)$, and the energy purchase cost from the main grid denoted by $\lambda_t \sum P_t^I$, where P_t^I denotes the electricity load at time t and λ_t is the electricity price at different times. Let P_t^G denote the electricity from the grid.

At each time $t \in T$, the station operation problem is described by a nonconvex problem:

$$\min_{P_t^D, P_t^C, P_t^I} \sum_{t \in T} f^D(P_t^D) - f^C(P_t^C) + \lambda_t \sum_{t \in T} P_t^I \quad (9)$$

Constraints:

$$P_t^C + P_t^I = P_t^D + P_t^G \quad (10)$$

$$0 \leq P_{i,t}^D \leq \bar{P}_i^D, \forall t \in T, \forall i \in N_{\text{batteries}} \quad (11)$$

$$0 \leq P_{i,t}^C \leq \bar{P}_i^C, \forall t \in T, \forall i \in N_{\text{batteries}} \quad (12)$$

$$\underline{E}_i \leq E_{i,t} \leq \bar{E}_i, \forall t \in T, \forall i \in N_{\text{batteries}} \quad (13)$$

$$E_{i,t-1} + \xi^C P_{i,t}^C - P_{i,t}^D / \xi^D = E_{i,t}, \forall t \in T, \forall i \in N_{\text{batteries}} \quad (14)$$

$$0 \leq P_t^I \leq \bar{P}_t^I, \forall t \in T \quad (15)$$

where f^C and f^D are the cost functions of charging and discharging based on real time prices, $P_{i,t}^C$ and $P_{i,t}^D$ are the charging and discharging power of battery i at time t , and they must be non-negative and no more than the maximum value. $E_{i,t}$ is the State of Charge (SOC) of station i , which cannot exceed the limits. ξ^C and ξ^D are the charging and discharging efficiency coefficients, P_t^I cannot exceed the station operation limits.

C. Inventory Level Model

Since the cost of a single battery is high, it is necessary to determine the inventory level of batteries at each BSW to improve the utilization rate and reduce the total cost [10]. Suppose that all BSWs obey the rule of Economic Order Quantity with Planned Backorder Model (EOQB) [11] to decide the battery inventory.

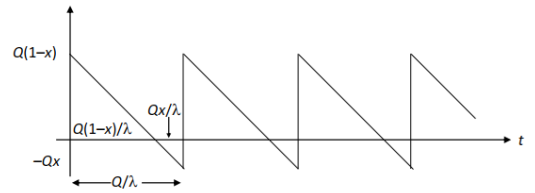


Fig. 3. The EOQB inventory curve.

The total average annual cost is:

$$g(Q, x) = \frac{K\lambda}{Q} + \frac{hQ(1-x)^2}{2} + \frac{pQx^2}{2}, \quad (16)$$

where, p indicates the backorder penalty per item per year; x is the fraction of backordered demand; K is the fixed cost per order; h is the inventory holding cost per unit per year; Q indicates the order quantity. Suppose that the demand is a deterministic constant. The rate λ satisfies $\lambda = Q/T_y$, where T_y is the order cycle. Computing partial derivatives of $g(Q, x)$ and setting them as zero,

$$\frac{\partial g}{\partial x} = -hQ(1-x) + pQx = 0 \quad (17)$$

$$\frac{\partial g}{\partial Q} = \frac{h(1-x)^2}{2} + \frac{px^2}{2} - \frac{K\lambda}{Q^2} = 0, \quad (18)$$

we can get $x^* = \frac{h}{h+p}$. Substituting x^* into $g(Q, x^*)$, then we can obtain:

$$Q^* = \sqrt{\frac{2K\lambda(h+p)}{hp}} \quad (19)$$

$$g(Q^*, x^*) = \sqrt{\frac{2K\lambda hp}{h+p}}. \quad (20)$$

Thus, we can obtain the best order quantity and the best cost per year per site, based on the parameters K, λ, h, p , collected and calculated from operation statistics.

IV. SOLUTION AND SIMULATION

To show the effectiveness of our proposed models, we construct a region map as shown in Fig. 6. The distribution of EV ownership in the region is simulated and the EV ownership is discounted and normalized by the ratio of charging demand to swapping demand. A bubble distribution diagram is then formed: the larger the purple bubble, the greater the demand at this node. The swapping service is limited by distance and demand, and the minimum coverage radius is 5km.

The decision variables included in the site selection model are x_{ij}, d_{ij}, y_i , and priority-based encoding of chromosomes. Suppose $I = \{1, 2, \dots, n\}$, the priority of warehouse i is denoted by $\rho_i \in I$ [8]. In this way, the chromosomes can be coded as follows by $\rho_1, \rho_2, \dots, \rho_n$. Then the top α genes are with the highest priority, and set $y_i = 1$ for the corresponding warehouse i , set $y_i = 0$ for the warehouse corresponding to the remaining $m - \alpha$ genes. For example, when $m = 9, \alpha = 4$, the priority-based encoding of chromosomes is shown in Fig. 4. The top four priorities of swapping facilities are $i = 2, 5, 7, 9$, so $y_i = 1, i = 2, 5, 7, 9$.

Site i	Gene:	1	2	3	4	5	6	7	8	9
Priority	Allele:	3	8	5	1	7	4	9	2	6

Fig. 4. Priority-based encoding of chromosomes.

For x_{ij} , a binary encoding is used and the string length is the number of candidate sites n . d_{ij} and y_i are both with floating point encoding and permutation encoding. The priority-based encoding can essentially be regarded as a permutation encoding. For this encoding, a variety of genetic operators have been investigated by previous studies. In the algorithm of this paper, the crossover uses the hybridization operator based on gene location. This method randomly extracts some offspring from one parent and then extracts genes heterozygous to them from another parent in order of the queue to fill the corresponding empty positions in turn to obtain the offspring genes as shown in Fig. 5.

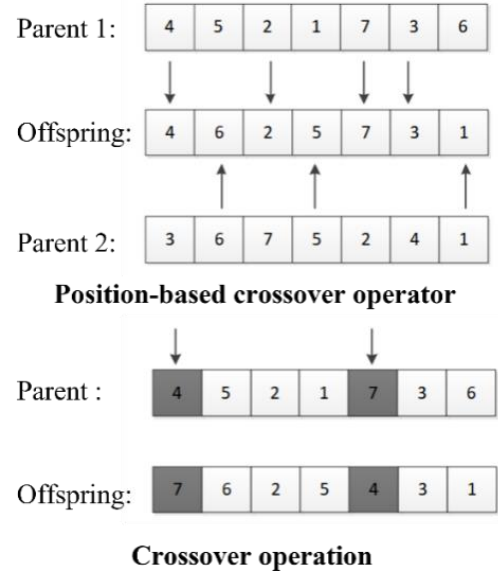


Fig. 5. Crossover operation.

The genetic parameters are determined as follows: after the mutation operation for the genes in the distribution center, new offspring are generated, assuming a mutation probability of 0.09 and a crossover probability of 0.48 [8]. The maximum number of iterations is set as 100, and the operation is terminated when the number is reached.

Totally, there are 24 candidate sites that meet the distance and demand constraints selected, as shown in the orange nodes in Fig. 6.

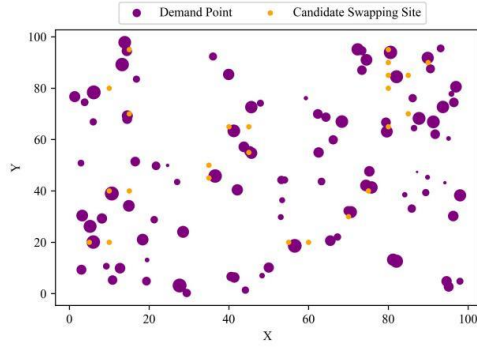


Fig. 6. The candidate swapping sites.

From the 24 selected candidate swapping station sites that already meet the minimum distance and demand constraints, select 6 sites to construct BSS/BSWs via the GA, and the service radius of each BSS/BSW is 20km. The sites selection results are shown in Fig. 7. The star in the middle of the map is a centralized charging and swapping station, and the remaining 5 points surrounded by red square are BSWs that are not connected to the grid.

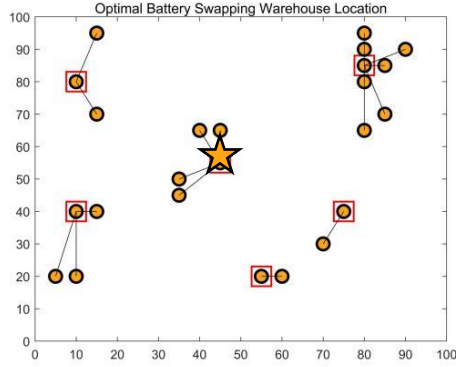


Fig. 7. Optimal battery swapping warehouse location.

For most charging stations, the peak load period is from 8-12a.m., with the fastest growth from 8-10a.m., and a large increase in load from 6-8p.m. If the power battery is chosen to charge at night during the low period, the charging cost of the swapping station will be reduced. Under the time-sharing tariff mechanism, the low-price period is from 22:00pm to 7:00am. Here we simulate 2 cases. In Case 1, there is only one CCS in the whole region, and all EVs with charging needs go to this station for charging, with the arrival distribution of EVs shown in Fig. 8. In case 2, there are both a CCS and BSWs in the region. Since the BSWs share the pressure of traffic flow, fewer vehicles will go to the CCS to recharge, and as a result, the queuing time here is reduced and the traffic congestion is relieved. Batteries unloaded at other BSWs are transported back to the CCS for charging, so used batteries can be uniformly charged starting at 22:00pm, which can reduce charging costs while charging as many batteries as possible. The current charging methods for power batteries are constant voltage charging, constant current charging, segmented charging, etc. Here the charging station chooses the more common constant current - constant voltage two-stage

charging method. The charging power varies with the charging current at the beginning of the charging stage, while the power changes very little at the end of the charging stage.

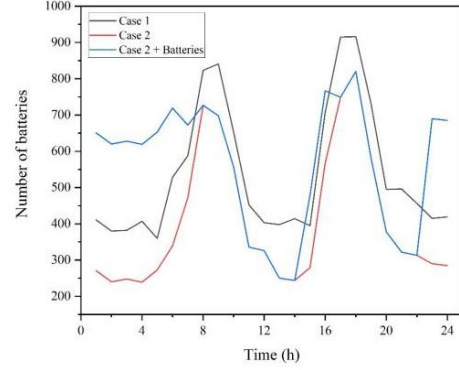


Fig. 8. Number of batteries to be charged at the center charging station.

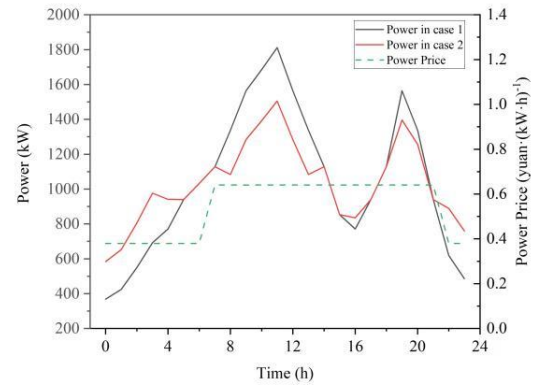


Fig. 9. The power load curves at the center charging station.

Set the station scheduling period to one day. In Case 1, all EVs arrive at the center station for replenishment according to the arrival law. In Case 2, the batteries unloaded at other BSWs are transported to the center station for replenishment during low peak periods. The electricity price curve and the load curve of CCS are shown in Fig. 9. It is obvious that the load curve of Case 1 is much steeper than the load curve of Case 2 during the peak period, and the batteries transported back from BSWs for recharging cause the load to increase during 22:00pm to 7:00am, filling in some of the valleys and allowing the station load curve to level out throughout the day.

V. CONCLUSION AND FUTURE WORK

In this paper, we have proposed a model for the layout of battery swapping stations combined with battery swapping warehouses. The purpose is to optimize the location layout and quantity of the battery swapping warehouse to increase the coverage of the service, increase the revenue, and cut the peak and fill the valley for the grid load. Since the power swapping warehouse is not connected to the grid, the batteries can be transported back to the charging station for charging during low-load periods. The bulk of the batteries are connected to the grid for replenishment during low peak periods, which has a more significant effect on the grid in terms of peak shaving and valley filling. Cause studies show

the effectiveness of the proposed method on increasing the service coverage, saving cost and increasing revenue while meeting more demands.

Future work will take into account traffic flow and path conditions for more accurate modeling of the locations, as well as the uncertainty of demand for more detailed modeling of warehouse inventory levels and battery inventory distribution in the region to further reduce costs.

REFERENCES

- [1] R. R. Kumar and K. Alok, "Adoption of electric vehicle: A literature review and prospects for sustainability," *Journal of Cleaner Production*, vol. 253, 119911, 2020.
- [2] J. Y. Yong, V. K. Ramachandaramurthy, K. M. Tan, and N. Mithulananthan, "A review on the state-of-the-art technologies of electric vehicle, its impacts and prospects," *Renewable and Sustainable Energy Reviews*, vol. 49, pp. 365-385, 2015.
- [3] Q. Kang, J. Wang, M. Zhou, and A. C. Ammari, "Centralized charging strategy and scheduling algorithm for electric vehicles under a battery swapping scenario," *IEEE Trans. Intelligent Transportation Systems*, vol. 17, no. 3, pp. 659-669, 2015.
- [4] H. Y. Mak, Y. Rong, and Z. J. M. Shen, "Infrastructure planning for electric vehicles with battery swapping," *Management Science*, vol. 59, no. 7, pp. 1557-1575, 2013.
- [5] L. He, H. Y. Mak, Y. Rong, and Z. J. M. Shen, "Service region design for urban electric vehicle sharing systems," *Manufacturing & Service Operations Management*, vol. 19, no. 2, pp. 309-327, 2017.
- [6] X. Tan, G. Qu, B. Sun, N. Li, and D. H. Tsang, "Optimal scheduling of battery charging station serving electric vehicles based on battery swapping," *IEEE Trans. Smart Grid*, vol. 10, no. 2, pp. 1372-1384, 2017.
- [7] T. Zhang, X. Chen, Z. Yu, X. Zhu, and D. Shi, "A Monte Carlo simulation approach to evaluate service capacities of EV charging and battery swapping stations," *IEEE Trans. Industrial Informatics*, vol. 14, no. 9, pp. 3914-3923, 2018.
- [8] J. Kratica, D. Tošić, V. Filipović, and I. Ljubić, "Solving the simple plant location problem by genetic algorithm," *RAIRO-Operations Research*, vol. 35, no. 1, pp. 127-142, 2001.
- [9] W. Wei, S. Mei, L. Wu, M. Shahidehpour, and Y. Fang, "Optimal traffic-power flow in urban electrified transportation networks," *IEEE Trans. Smart Grid*, vol. 8, no. 1, pp. 84-95, 2017.
- [10] M. T. Melo, S. Nickel, and F. Saldanha-Da-Gama, "Facility location and supply chain management—A review," *European Journal of Operational Research*, vol. 196, no. 2, pp. 401-412, 2009.
- [11] G. A. Widyadana, L. E. Cárdenas-Barrón, and H. M. Wee, "Economic order quantity model for deteriorating items with planned backorder level," *Mathematical and Computer Modelling*, vol. 54, no. 5-6, pp. 1569-1575, 2011.